

YEHYA FARHAT

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Research Interests: Machine Learning | Convex & Non-Convex Optimization | Agentic AI | AI for Science & Engineering

EDUCATION

Rice University Aug 2024 - May 2028
PhD in Computer Science Houston, TX

- GPA: 4.0/4.0
- Advisor: [Prof. Anastasios Kyrillidis](#)

Syracuse University Aug 2021 - May 2023
Master of Science in Computer Science Syracuse, NY

American University of Beirut Aug 2016 - Dec 2020
Bachelor of Science in Computer Science, Minor in Mathematics Beirut, LB

PUBLICATIONS & PREPRINTS

Farhat Y.*, Shili H.*, Liao F.*, Dun C.*, Garcia M.*, Zheng G., Awadallah A., Sim R., Dimitriadis D., Kyrillidis A.
"Learning to Specialize: Joint Gating-Expert Training for Adaptive MoEs in Decentralized Settings" [🔗](#)
Conference on Neural Information Processing Systems; Neurips-2025

Farhat Y.
"On the Robustness of Decision-Focused Learning" [🔗](#)

Farhat Y.
"End-to-End Decision-Focused Learning Using Learned Solvers" [🔗](#)

WORK EXPERIENCE

Data Scientist June 2022 – August 2022
Qnovo Milpitas, CA

- Operated as a full-stack engineer to design and build a battery testing platform and a corresponding user interface to query the database, reduced time spent on ad-hoc reporting from 2-3 days to 10 minutes
- Cleaned and performed exploratory data analysis on 5M + rows of time-series data. Analysis was used to confirm the existence of a sensitivity problem in battery capacity estimates

Software Engineer April 2021 – July 2021
Precast FZCO Dubai, UAE

- Devised a cloud-based micro-architecture and employed Microsoft Azure to establish a correspondence between CMMS software and the client's IoT software, thereby streamlining task creation between software systems and reducing task creation time to zero
- Designed and implemented an integration module to connect multiple ERP, IoT, and CMMS software systems. The module included an interface allowing clients to specify what information to streamline between software systems

RESEARCH EXPERIENCE

Research Assistant August 2024 – Current
Rice University
Advisor: [Prof. Anastasios Kyrillidis](#)

Research Scholar September 2023 – June 2024
Syracuse University
Advisor: [Prof. Venkata Gandikota](#)

- Studied the expressiveness of piecewise linear functions attained by deep neural networks through the lens of linear regions. Work involves analyzing the number of regions attained by the learned functions, with a focus on the impact of various loss functions, optimization algorithms, and training cycles.
- Studied extensions for a distributed Quasi-Newton method for general first-order optimization

Research Assistant (Master's Thesis)

Syracuse University

Advisor: [Prof. Ferdinando Fioretto](#)

August 2022 – May 2023

- Conducted a survey of End-to-End constrained optimization learning and researched an amortized method for the Decision-Focused Learning paradigm. Our framework involves developing a surrogate machine learning model for a continuous relaxation of the discrete optimization problem and substituting the learned model with the optimization solver within the pipeline.
- ["End-to-End Decision-Focused Learning Using Learned Solvers"](#)  (2023). Theses - ALL. 714. <https://surface.syr.edu/thesis/714>

PROJECTS

Transformers for Higher-Order Logic

May 2022

- Collaborated with a small team to implement attention-based mechanisms for learning higher-order logic theorem proving. We implemented multiple language models and trained them on the HolStep dataset in an attempt to study automatic theorem proving; [Project Link](#) 

Optimal Stopping Problem with Unknown Variable

December 2021

- Analyzed one form of the optimal stopping problem known as the secretary problem which has a closed form solution. The equation requires the number of candidates to be known beforehand. Collaborated with a team member to attempt to find an efficient solution to the same problem but without access the number of candidates; [Project Link](#) 

University Petition System

December 2020

- Collaborated with three team members to develop a course petition website for the American University of Beirut. This platform enables students to submit their petitions along with their transcripts, which are automatically parsed. The chairperson is notified if the students meet the petition requirements.
- This project enabled us to significantly streamline the process, saving the chairperson countless hours previously spent organizing and reviewing transcripts to determine if the requirements have been met;
- Technology stack: JavaScript | Node.js | React | MongoDB
[Project Link](#) 

Power Monitor

July 2020

- Cofounded & developed power monitor, pitched to Lebanon's public government energy sector. Enable the sector to transition into digital bills and cut down on the required transportation. Saving the government thousands of dollars each month
- Allows users to track daily/monthly usage, electricity bill, and carbon footprint using the phone application; [Product Website](#) 

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Julia, Dart

ML/DS: PyTorch, Numpy, Keras, SciPy, CvxPy, Scikit-Learn

Other: Flask, React, Flutter, SQL, GraphQL, REST API, Selenium, CUDA, HTCondor, Azure, AWS

FELLOWSHIP & AWARDS

Graduate Student Travel Grant 2024-2025

Travel award from Rice Engineering Alumni to support engineering graduate students who are attending conferences or professional meetings to present their research work.

TEACHING EXPERIENCE

Graduate Design and Analysis of Algorithms

Teaching Assistant

COMP 582 | Rice University

Spring 2025

Houston, TX

- Graduate course: Advanced Design and Analysis of algorithms.
Taught by [Prof. Xiaoyun Fu](#)

Introduction to Object-Oriented Design

Teaching Assistant
CSE 283 | Syracuse University

Spring 2023
Syracuse, NY

- Undergraduate course: Fundamental software design concepts of functional decomposition and object-oriented design. Taught by [Prof. Farzana Rahman](#)

Computer Organization & Programming Systems

Teaching Assistant
CIS 341 | Syracuse University

Spring 2023
Syracuse, NY

- Undergraduate course: Essentials of computer organization, digital logic, microprogramming, processors, memories, input-output devices. Instruction sets, assembly language, data types, relation to high level languages. Taught by [Prof. Farzana Rahman](#)

Assured Programming and Formal Methods

Teaching Assistant
CIS 623 | Syracuse University

Fall 2022
Syracuse, NY

- Graduate-level core course: Reasoning about programs through the specification, design, and realization of provably correct programs. Taught by [Prof. Andrew Lee](#)

Advanced Computer Architecture

Teaching Assistant
CIS 655/CSE 661 | Syracuse University

Spring 2022
Syracuse, NY

- Graduate-level core course: Memory systems, pipelining, simultaneous multithreading, run-time optimization, array processing, parallel processing, multiprocessing. Taught by [Prof. Ehat Ercanli](#)

OUTREACH & EXTRACURRICULAR

President and Co-Founder

Syracuse University Graduate Chess Club

February 2022 – October 2022
Syracuse, NY

- Co-founded the Graduate Chess Club at Syracuse University and served as the Acting President, orchestrating the club's growth with over 150 participants during my tenure. Successfully merged the meetings of the graduate and undergraduate chess clubs, fostering a cohesive and inclusive chess community within Syracuse University.

Math and Physics Tutor Volunteer

MMKN

January 2018 – July 2019
Beirut, Lebanon

- Served as a part-time Physics and Mathematics teacher for students in grades 9 and 10. Specifically, I conducted afternoon classes catering to refugees and members of minority communities in Lebanon.

Blog Posts

- [How to Build Your Own Custom Optimizer in PyTorch \(Step-by-Step\)](#)
Farhat, Y. (2025, June). How to Build Your Own Custom Optimizer in PyTorch (Step-by-Step). Medium.
- [The Most Fundamental Unsolved Problem in All of Science and Philosophy](#)
Farhat, Y. (2023, December). The Most Fundamental Unsolved Problem in All of Science and Philosophy. Medium.
- [What Are the Machines Learning?](#)
Farhat, Y. (2023, September). What Are the Machines Learning?